# QuiverCombinatoricsTools

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# CONTENTS

*QuiverCombinatoricsTools* is a SageMath package that adds combinatorial functions to *QuiverTools* to calculate symplectic leaves of quiver varieties, available here https://github.com/emanuel-roth/QuiverCombinatoricsTools. It adds on to the *QuiverTools* package written by Pieter Belmans, Hans Franzen and Gianni Petrella, as seen here https://sage.quiver.tools/ and here https://github.com/QuiverTools/QuiverTools, so consult their documentation when needed.

To install it, make sure you have both *QuiverTools* and *QuiverCombinatoricsTools*

```
sage --pip install git+https://github.com/QuiverTools/QuiverTools.git
sage --pip install git+https://github.com/emanuel-roth/QuiverCombinatoricsTools.git
```

and then you can simply run

```
from quiver import *
from quivercombinatorics import *
```

to get started.

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# CHAPTER ONE

# GENERATING QUIVERS

Here are some functions that help generate quivers to test examples.

`quivercombinatorics.`**`quiver_from_cartan_matrix`**`(`*$C$*`)`

Returns the quiver $Q$ given by a Cartan matrix $C$

INPUT:

- `C` – a square matrix with entries in $\mathbb{Z}$

OUTPUT: The quiver $Q$ given by a Cartan matrix $C$

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: C = [[2, -1, 0], [-1, 2, -2], [0, -2, 2]]
sage: quiver_from_cartan_matrix(C)
a quiver with 3 vertices and 3 arrows
```

`quivercombinatorics.`**`random_quiver`**`(`*vertices*, *max_arrows_per_edge*`)`

Returns a randomly generated quiver

INPUT:

- `vertices` – number of vertices you want in the quiver, in $\mathbb{N}_0$
- `max_arrows_per_edge` – maximum of number of arrows you want in the quiver, in $\mathbb{N}_0$

OUTPUT: A randomly generated quiver with the prescribed vertices and maximum number of edges

EXAMPLES:

```
sage: from quivercombinatorics import *
sage: random_quiver(3, 2)
a quiver with 3 vertices and 6 arrows

sage: from quivercombinatorics import *
sage: random_quiver(46, 25)
a quiver with 46 vertices and 26388 arrows
```

We notate quivers by $Q$, with vertices in $Q_0$ and edges in $Q_1$.

CHAPTER

TWO

# CONSTRUCTING $\Sigma_\lambda$

We follow William Crawley-Boevey in this paper. Given a quiver $Q$ with $n$ vertices, and $\lambda \in \mathbb{Z}^n$, $\Sigma_\lambda$ is the set of $\alpha \in \mathbb{N}^n$ such that $\alpha$ is a positive root of $Q$, $\alpha \cdot \lambda = 0$, and

$$p(\alpha) > \sum_{t=1}^{r} p(\beta^{(t)})$$

for any decomposition $\alpha = \beta^{(1)} + \cdots + \beta^{(r)}$ with $r \geq 2$ and $\beta^{(t)}$ a positive root of $Q$ with $\lambda \cdot \beta^{(t)} = 0$ for all $t$.

`Quiver.`**`p_function`**`(`$x$`)`

Outputs the function $p(x) = 1 - \frac{1}{2}(x, x)$, where $(x, x)$ is the symmetrized Euler form. Note that $p(x) \geq 0$ if $x$ is a root, and $p(x) = 0$ if and only if $x$ is a real root

INPUT:

- `x` – an element of $\mathbb{Z}Q_0$

OUTPUT: $1 - \frac{1}{2}(x, x)$

EXAMPLE:

The $p$ function evaluated at $(2, 3)$ of the doubled A2 quiver:

```
sage: from quivercombinatorics import *
sage: Q = Quiver([[0, 1], [1, 0]])
sage: Q.p_function((2, 3))
0.0
```

We write $R_\lambda^+$ to denote the set of positive roots $\alpha$ with $\alpha \cdot \lambda = 0$ and $\mathbb{N}R_\lambda^+$ for the set of sums of elements of $R_\lambda^+$. Using Theorem 5.6 of Crawley-Boevey's paper.

**Theorem (Crawley-Boevey, 2001)**

For $\alpha \in \mathbb{N}^n$, then $\alpha \in \Sigma_\lambda$ if and only if $0 \neq \alpha \in \mathbb{N}R_\lambda^+$ and $(\beta, \alpha - \beta) \leq -2$ whenever $\beta \in \mathbb{N}^n$ and $0 < \alpha < \beta$.

`Quiver.`**`R_lambda_plus`**`(`$l$, $v$`)`

Returns a list of elements of $R_\lambda^+$, the set of positive roots $\alpha$ with $\alpha \cdot \lambda = 0$, up to the upper bound $v$

INPUT:

- `l` – an element of $\mathbb{Z}Q_0$
- `v` – an element of $\mathbb{N}Q_0$

OUTPUT: A list of elements of $R^+_\lambda$, where `l` is $\lambda$, the set of positive roots $\alpha$ with $\alpha \cdot \lambda = 0$, up to the upper bound $v$

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: Q = Quiver([[0, 1], [1, 0]])
sage: Q.R_lambda_plus((1, -1), (5, 5))
[(1, 1), (2, 2), (3, 3), (4, 4)]
```

We also need the helper function `N_set`.

`quivercombinatorics.`**`N_set`**`(S, v)`

For a list of vectors $S$ in $\mathbb{Z}Q_0$, returns all possible sums of vectors in $S$ as a list, up to the upper bound $v$. Though not directly about quivers, this is a helper function to define $\Sigma_\lambda$

INPUT:

- `S` – a list of vectors $S$ in $\mathbb{Z}Q_0$
- `v` – vector in $\mathbb{Z}Q_0$

OUTPUT: A list of all possible sums of vectors in $S$ as a list, up to the upper bound $v$

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: N_set([[0, 1]],[0, 3])
[(0, 1), (0, 2), (0, 3)]
```

With these functions, we can define $\Sigma_\lambda$.

`Quiver.`**`sigma_lambda`**`(l, v)`

Returns a list of elements of $\Sigma_\lambda$, up to the upper bound $v$

INPUT:

- `l` – an element of $\mathbb{Z}Q_0$
- `v` – an element of $\mathbb{N}Q_0$

OUTPUT: A list of elements of $\Sigma_\lambda$, where `l` is $\lambda$, up to the upper bound $v$

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: Q = Quiver([[0, 1], [1, 0]])
sage: Q.sigma_lambda((1, -1), (5, 5))
[(1, 1)]
```

CHAPTER

THREE

# CB-DECOMPOSITIONS

A *representation type* of $x$ of a quiver $Q$ is a sequence $\tau = (\beta^{(1)}, n_1; \ldots; \beta^{(k)}, n_k)$ with $\beta^{(i)} \in \Sigma_\lambda$ and $n_i \in \mathbb{N}$ such that

$$x = \sum_{i=1}^{k} n_i \beta^{(i)}$$

We allow $\beta^{(i)}$ to occur multiple times if it is an imaginary root, i.e. $p(\beta^{(i)}) > 0$. The *CB-decomposition* of $x$ is defined to be the unique representation type of $x$ for which

$$p(\tau) := \sum_{i=1}^{k} p(\beta^{(i)})$$

is maximal. In order to construct CB-decompositions (called canonical decompositions in this paper, but are *not* the same as `canonical_decomposition` from *QuiverTools*). We first need to find all representation types of $x$, up to a bound $v$, for which we need the following helper functions.

`quivercombinatorics.`**`vector_decomposition`**`(x, S)`

For a list of vectors $S$ in $\mathbb{Z}Q_0$, returns all possible sums of vectors in $S$ as a list that sum up to $x$. This is a helper function in order to define CB-decompositions

INPUT:

- `x` – a vector in $\mathbb{Z}Q_0$
- `S` – a list of vectors $S$ in $\mathbb{Z}Q_0$

OUTPUT: All possible sums of vectors in $S$ as a list that sum up to $x$

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: vector_decomposition((4, 5), [(0, 1),(1, 0),(1, 1)])
[[[(0, 1), 1], [(1, 1), 4]],
[[(0, 1), 2], [(1, 0), 1], [(1, 1), 3]],
[[(0, 1), 3], [(1, 0), 2], [(1, 1), 2]],
[[(0, 1), 4], [(1, 0), 3], [(1, 1), 1]],
[[(0, 1), 5], [(1, 0), 4]]]
```

`quivercombinatorics.`**`small_decomposition`**`(v, n)`

For a vector $v$ in $\mathbb{Z}Q_0$, it returns all possible partitions of the representation type $[v, n]$

INPUT:

- `v` – a vector in $\mathbb{Z}Q_0$
- `n` – a natural number in $\mathbb{N}$

OUTPUT: All possible partitions of the representation type $[v, n]$

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: small_decomposition((1, 1), 3)
[[[(1, 1), 1], [(1, 1), 1], [(1, 1), 1]],
[[(1, 1), 2], [(1, 1), 1]],
[[(1, 1), 3]]]
```

With these functions, we can determine all representation types.

`Quiver.`**`all_representation_types`**`(`$l, x$`)`

Returns a list of representation types of a quiver, with respect to $x$

INPUT:

- `l` – an element of $\mathbb{Z}Q_0$
- `x` – an element of $\mathbb{N}Q_0$

OUTPUT: A list of representation types of a quiver, with respect to $x$, and where `l` is $\lambda$. Each representation type is stored as a list whose elements are 2-tuples, the first element is in $\mathbb{N}Q_0$, and the second is in $\mathbb{N}$

EXAMPLES:

```
Q = Quiver([[0, 1], [1, 0]])
Q.all_representation_types((1, -1), (5, 5))
[[[(1, 1), 1], [(1, 1), 1], [(1, 1), 1], [(1, 1), 1], [(1, 1), 1]],
[[(1, 1), 1], [(1, 1), 1], [(1, 1), 1], [(1, 1), 2]],
[[(1, 1), 1], [(1, 1), 1], [(1, 1), 3]],
[[(1, 1), 1], [(1, 1), 2], [(1, 1), 2]],
[[(1, 1), 1], [(1, 1), 4]],
[[(1, 1), 2], [(1, 1), 3]],
[[(1, 1), 5]]]

sage: C = [[2,-1,0], [-1,2,-2], [0,-2,2]]
sage: Q = quiver_from_cartan_matrix(C)
sage: Q.all_representation_types((0,0,0), (2,4,3))
[[[(0, 0, 1), 1],
[(0, 1, 0), 2],
[(0, 1, 1), 1],
[(0, 1, 1), 1],
[(1, 0, 0), 2]],
[[(0, 0, 1), 1], [(0, 1, 0), 2], [(0, 1, 1), 2], [(1, 0, 0), 2]],
[[(0, 0, 1), 2], [(0, 1, 0), 3], [(0, 1, 1), 1], [(1, 0, 0), 2]],
[[(0, 0, 1), 3], [(0, 1, 0), 4], [(1, 0, 0), 2]],
[[(0, 1, 0), 1],
[(0, 1, 1), 1],
[(0, 1, 1), 1],
[(0, 1, 1), 1],
[(1, 0, 0), 2]],
[[(0, 1, 0), 1], [(0, 1, 1), 1], [(0, 1, 1), 2], [(1, 0, 0), 2]],
[[(0, 1, 0), 1], [(0, 1, 1), 3], [(1, 0, 0), 2]],
[[(2, 4, 3), 1]]]
```

Representation types of $x$ correspond to a symplectic leaf of the quiver variety $M_0(Q, x)$, and the symplectic leaf dimension is $2p$ applied to the representation type, where $p$ is the `p_function`.

Quiver.**symplectic_leaf_dimension**(*tau*)

Returns the dimension of the symplectic leaf corresponding to the representation type $\tau$

INPUT:

- tau – a representation type, i.e., a list whose elements are 2-tuples, the first element is in $\mathbb{N}Q_0$, and the second is in $\mathbb{N}$

OUTPUT: The corresponding symplectic leaf dimension, i.e., $2p$ applied to every element of $\tau$

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: Q = Quiver([[0, 1],[1, 0]])
sage: Q.symplectic_leaf_dimension([[(1, 1), 2], [(1, 1), 3]])
4
```

Quiver.**CB_decomposition**($l$, $x$)

Returns the CB decomposition, which is the representation type maximizing $p$

INPUT:

- l – an element of $\mathbb{Z}Q_0$
- x – an element of $\mathbb{N}Q_0$

OUTPUT: The CB decomposition with respect to $x$, where l is $\lambda$

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: Q = Quiver([[0, 1],[1, 0]])
sage: Q.CB_decomposition((1, -1), (5, 5))
[[(1, 1), 1], [(1, 1), 1], [(1, 1), 1], [(1, 1), 1], [(1, 1), 1]]
```

Knowing the CB-decomposition, it is then easy to calculate the dimension of the quiver variety $M_0(Q, x)$, as it gives the largest (open) symplectic leaf inside $M_0(Q, x)$.

Quiver.**quiver_variety_dimension**($l$, $x$)

Returns the dimension of the quiver variety

INPUT:

- l – an element of $\mathbb{Z}Q_0$
- x – an element of $\mathbb{N}Q_0$

OUTPUT: The dimension of the corresponding quiver variety

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: Q = Quiver([[0, 1],[1, 0]])
sage: Q.quiver_variety_dimension((1, -1), (5, 5))
10
```

It's often important to know the codimension 2 leaves of the quiver variety $M_0(Q, x)$, so we code this here.

Quiver.**codimension_two_leaves**($l$, $x$)

Returns the codimension 2 leaves of the quiver variety

INPUT:

- `l` – an element of $\mathbb{Z}Q_0$
- `x` – an element of $\mathbb{N}Q_0$

OUTPUT: The representation types of the codimension 2 leaves of the quiver variety

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: Q = Quiver([[0, 1],[1, 0]])
sage: Q.codimension_two_leaves((0, 0), (5, 5))
[[[(0, 1), 1],
[(1, 0), 1],
[(1, 1), 1],
[(1, 1), 1],
[(1, 1), 1],
[(1, 1), 1]],
[[(1, 1), 1], [(1, 1), 1], [(1, 1), 1], [(1, 1), 2]]]
```

# CHAPTER FOUR

# ext-QUIVERS

Let $\tau = (\beta^{(1)}, n_1; \ldots; \beta^{(k)}, n_k)$ be a representation type of $Q$. The ext-*quiver* $\tilde{Q}$ associated to the symplectic leaf is the quiver with $k$ vertices where the number of edges between vertex $i$ and vertex $j$ is $-(\beta^{(i)}, \beta^{(j)})$ and the number of loops at vertex $i$ is $p(\beta^{(i)})$. The corresponding dimension vector of $\tilde{Q}$ is $\mathbf{n} = (n_1, \ldots, n_k)$, and we take $\tilde{\lambda} = 0$.

`Quiver.`**`ext_quiver`**`(`*tau*`)`

Given a representation type, returns the ext-quiver

INPUT:

- `tau` – a representation type, i.e., a list whose elements are 2-tuples, the first element is in $\mathbb{N}Q_0$, and the second is in $\mathbb{N}$

OUTPUT: The ext-quiver

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: Q = Quiver([[0, 1], [1, 0]])
sage: tau = [[(1, 1), 2], [(1, 1), 1], [(1, 1), 1], [(1, 1), 1]]
sage: Q.ext_quiver(tau)
a quiver with 4 vertices and 4 arrows
```

`quivercombinatorics.`**`ext_dimension_vector`**`(`*tau*`)`

For a representation type $\tau$, returns the associated dimension vector

INPUT:

- `tau` – a representation type, i.e., a list whose elements are 2-tuples, the first element is in $\mathbb{N}Q_0$, and the second is in $\mathbb{N}$

OUTPUT: a vector in $\mathbb{N}Q_0$

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: tau = [[(1, 1), 2], [(1, 1), 1], [(1, 1), 1], [(1, 1), 1]]
sage: ext_dimension_vector(tau)
(2, 1, 1, 1)
```

CHAPTER

FIVE

# CLASSIFICATION OF MINIMAL DEGENERATIONS OF SYMPLECTIC LEAVES

For a quiver $Q$, a positive root $\alpha$ is *minimal imaginary* if it is imaginary and given any positive root $\beta$, $\alpha > \beta$ implies $\beta$ is real.

`Quiver.`**`is_minimal_imaginary_root`**`(x)`

Tests whether a vector is a vector is a minimal imaginary root

INPUT:

- `x` – a vector in $\mathbb{Z}Q_0$

OUTPUT: `True` if `x` is a minimal imaginary root, and `False` otherwise

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: Q = KroneckerQuiver(2)
sage: Q.is_minimal_imaginary_root((1,1))
True
```

`Quiver.`**`all_minimal_imaginary_positive_roots`**`(v)`

Returns all minimal imaginary positive roots up to a bound $v$

INPUT:

- `v` – a vector in $\mathbb{N}Q_0$

OUTPUT: A list of all minimal imaginary positive roots up to a bound $v$

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: Q = KroneckerQuiver(2)
sage: Q.all_minimal_imaginary_positive_roots((3,3))
[(1,1)]
```

Given a quiver $Q$, a *subquiver* $T = (T_0, T_1)$ is specified by a subset $T_0 \subset Q_0$ of the vertices. Then, for $i, j \in T_0 \subset Q_0$ the number of edges between vertex $i$ and $j$ in $T$ equals the number of edges between vertex $i$ and $j$ in $Q$. In particular, if $i \in T_0$ then the number of loops at $i$ in $T$ is the same as the number of loops at $i$ in $Q$. The support of a dimension vector $v$ is the subset $T_0 \subset Q_0$ of all vertices $i$ such that $v_i \neq 0$. We think of the support of $v$ as a subquiver of $Q$.

We say that the subquiver $T$ is of *affine ADE type* if it is one of the graphs appearing in the classification of simply-laced affine Dynkin diagrams $\tilde{A}_l$, $\tilde{D}_l$ or $\tilde{E}_6$, $\tilde{E}_7$, $\tilde{E}_8$. Then there is a unique imaginary root $\delta$, with $p(\delta) = 1$, whose support is $T$. The key result is the classification of minimal imaginary roots is the following (Bellamy-Schedler 2026).

**Theorem (Bellamy-Schedler, 2026)**

If $M$ is a closed leaf in $M_0(Q, x)$ and $M \subset \bar{L}$ is a minimal degeneration, then $\bar{L} \cong M \times S$, where $S$ is isomorphic to one of the following isolated singularities:

1. A Kleinian singularity $(\mathbb{C}^2/\Gamma, 0)$.
2. The type A minimal nilpotent orbit closure $(\mathcal{O}_{\text{min}}, 0)$ in $\mathfrak{sl}_r(\mathbb{C})$.
3. $(\mathbb{C}^{2g}/\mathbb{Z}_2, 0)$.
4. Spec$(\mathbb{C}[x \mid \deg x \geq 2])$.

We define $\tilde{\tau}(\beta, n) := (\beta, n; e_1, \alpha_1 - n\beta_1; \dots; e_r, \alpha_r - n\beta_r)$. In each of the above cases, the leaf $L$ from this theorem corresponds to representation types:

1. $\tilde{\tau}(\delta, n)$ for $\delta$ a minimal imaginary root on an affine ADE subquiver.
2. $\tilde{\tau}((1, 1), n)$ for a two-vertex subquiver with $t \geq 3$ edges between the vertices and no loops at the vertices.
3. $\tau = (e_i, a; e_i, a; e_1, \alpha_1; \dots; e_{i-1}, \alpha_{i-1}; e_{i+1}, \alpha_{i+1}; \dots)$ where $i$ is a vertex with $g \geq 1$ loops and $2a = \alpha_i$.
4. $\tau = (e_i, a; e_i, b; e_1, \alpha_1; \dots; e_{i-1}, \alpha_{i-1}; e_{i+1}, \alpha_{i+1}; \dots)$ where $i$ is a vertex with $g \geq 1$ loops and $0 < a \neq b < \alpha_i$ with $a + b = \alpha_i$.

We assign the following labels to each subminimal representation type (i.e., representation types corresponding to a minimal degeneration):

1. $A_l, D_l$ or $E_6, E_7, E_8$ for affine Dynkin diagram of type $\tilde{A}_l, \tilde{D}_l$ or $\tilde{E}_6, \tilde{E}_7, \tilde{E}_8$.
2. $a_{r-1}$.
3. $c_g$.
4. $m_g$.

`Quiver.`**`all_subminimal_representation_types`**`(v)`

Returns all subminimal representation types up to a bound $v$, and classifies them by affine Dynkin type, or $a_{r-1}$, $c_g$, $m_g$

INPUT:

- `v` – a vector in $\mathbb{N}Q_0$

OUTPUT: A list of 2-tuples, the first element is a subminimal representation type up to a bound $v$, the second element is its corresponding affine Dynkin type, or $a_{r-1}$, $c_g$, $m_g$

EXAMPLES:

```
sage: from quivercombinatorics import *
sage: Q = CyclicQuiver(3)
sage: Q.all_subminimal_representation_types((3, 3, 3))
[([[(0, 0, 1), 2], [(0, 1, 0), 2], [(1, 0, 0), 2], [(1, 1, 1), 1]], 'A_{2}'),
([[(0, 0, 1), 1], [(0, 1, 0), 1], [(1, 0, 0), 1], [(1, 1, 1), 2]], 'A_{2}'),
([[(1, 1, 1), 3]], 'A_{2}')]

sage: from quivercombinatorics import *
sage: Q = LoopQuiver(3)
sage: Q.all_minimal_non_trivial_representation_types((5))
[([[(1), 1], [(1), 4]], 'm_{3}'), ([[(1), 2], [(1), 3]], 'm_{3}')]
```

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```
sage: from quivercombinatorics import *
sage: A = [[0, 1, 0, 3], [1, 0, 0, 0], [0, 0, 2, 0], [0, 0, 0, 0]]
sage: Q = Quiver(A)
sage: Q.all_minimal_non_trivial_representation_types((1, 1, 4, 1))
[([[(0, 0, 0, 1), 1], [(0, 0, 1, 0), 4], [(1, 1, 0, 0), 1]], 'A_1'),
([[(0, 0, 1, 0), 4], [(0, 1, 0, 0), 1], [(1, 0, 0, 1), 1]], 'a_{2}'),
([[(0, 0, 0, 1), 1],
[(0, 0, 1, 0), 1],
[(0, 0, 1, 0), 3],
[(0, 1, 0, 0), 1],
[(1, 0, 0, 0), 1]],
'm_{2}'),
([[(0, 0, 0, 1), 1],
[(0, 0, 1, 0), 2],
[(0, 0, 1, 0), 2],
[(0, 1, 0, 0), 1],
[(1, 0, 0, 0), 1]],
'c_{2}')]
```

CHAPTER

SIX

# PLOTTING SYMPLECTIC LEAVES IN HASSE DIAGRAMS

We visualize the poset of symplectic leaves (equivalently of representation types) as a Hasse diagram: The vertices in this diagram are the representation types and the minimal degenerations, represented by an edge connecting two representation types, correspond to the situation where $L_1 \leq L_2$ (equivalently, $\tau_1 \leq \tau_2$) but there does not exist $L_3$ such that $L_1 < L_3 < L_2$. We wish to compute the Hasse diagram, for which there are two methods.

**Method 1**

Traverse the Hasse diagram from the bottom, starting with the lowest leaves. When we reach a leaf $L_i$, we compute the ext-quiver associated to $L_i$. On this ext-quiver, we compute subminimal representation types of **n**. For each such representation type $\tilde{\tau} = (\gamma^{(1)}, m_1; \dots; \gamma^{(r)}, m_r)$, define:

$$D(\tilde{\tau}) := \left(D(\gamma^{(1)}), m_1; \dots; D(\gamma^{(r)}), m_r\right)$$

By (Bellamy-Schedler 2026), it is known that $D(\tilde{\tau})$ is a representation type for $v$.

In this way, we have a rule that assigns to each subminimal representation type of $\tilde{Q}$ a representation type $D(\tilde{\tau}(\alpha))$ of $v$. It is known that $\tau < D(\tilde{\tau}(\alpha))$ is a minimal degeneration and they all occur in this way. Proceeding in this way allows one to build the Hasse diagram from the bottom up.

`quivercombinatorics.`**`D_map`**`(`*m*, *tau*`)`

Evaluates a map $D : \mathbb{Z}^k \to \mathbb{Z}^n$ from dimension vectors of the ext-quiver to dimension vectors of the original quiver by $D(m_1, \dots, m_k) := \sum_{i=1}^{k} m_i \beta^{(i)}$

INPUT:

- `m` – a vector in $\mathbb{Z}\,\mathrm{ext}(Q)_0$
- `tau` – a representation type of the original quiver, i.e., a list whose elements are 2-tuples, the first element is in $\mathbb{N}Q_0$, and the second is in $\mathbb{N}$

OUTPUT: a vector in $\mathbb{Z}Q_0$

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: tau = [[(1, 1), 2], [(1, 1), 1], [(1, 1), 1], [(1, 1), 1]]
sage: D_map((1, 2, 8, -3), tau)
(8, 8)
```

`quivercombinatorics.`**`D_map_on_rep`**`(`*tau*, *L*`)`

Applies `d_map` to a representation type of the ext-quiver

INPUT:

- `L` – a representation type of the ext-quiver, i.e., a list whose elements are 2-tuples, the first element is in $\mathbb{N}\mathrm{ext}(Q)_0$, and the second is in $\mathbb{N}$
- `tau` – a representation type, i.e., a list whose elements are 2-tuples, the first element is in $\mathbb{N}Q_0$, and the second is in $\mathbb{N}$

OUTPUT: a representation type of the ext-quiver

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: tau = [[(1, 1), 2], [(1, 1), 1], [(1, 1), 1], [(1, 1), 1]]
sage: D_map_on_rep([[(1, 2, 8, -3), 3]],tau)
[[(8, 8), 3]]
```

`Quiver.`**`minimal_degenerations`**($L$)

Returns all minimal degenerations of a a symplectic leaf, corresponding to the representation type $L$

INPUT:

- `L` – a representation type, i.e., a list whose elements are 2-tuples, the first element is in $\mathbb{N}Q_0$, and the second is in $\mathbb{N}$

OUTPUT: A list of representation types

EXAMPLES:

```
sage: from quivercombinatorics import *
sage: A = [[0, 1, 0, 3], [1, 0, 0, 0], [0, 0, 2, 0], [0, 0, 0, 0]]
sage: Q = Quiver(A)
sage: Q.minimal_degenerations([[(0, 0, 0, 1), 1], [(0, 0, 1, 0), 4], [(1, 1, 0,␣
↪0), 1]])
[([[(0, 0, 1, 0), 4], [(1, 1, 0, 1), 1]], '$a_{2}(1)$'),
([[(0, 0, 0, 1), 1], [(0, 0, 1, 0), 1], [(0, 0, 1, 0), 3], [(1, 1, 0, 0), 1]],
'$m_{2}(1)$'),
([[(0, 0, 0, 1), 1], [(0, 0, 1, 0), 2], [(0, 0, 1, 0), 2], [(1, 1, 0, 0), 1]],
'$c_{2}(1)$')]
```

`Quiver.`**`get_Hasse_diagram_method_1`**(*l*, *v*, *dimensions=False*)

Applies Method 1 to obtain data for the Hasse diagram of minimal degenerations

INPUT:

- `l` – an element of $\mathbb{Z}Q_0$
- `v` – an element of $\mathbb{N}Q_0$
- `dimensions` – When set to `True`, the leaves will be labeled by (`symplectic_leaf_dimension`,``counter``), where `counter` enumerates symplectic leaves of equal dimensions

OUTPUT: `all_leaves`, `elements`, `relations`, `edge_labels`, the latter three are required to construct the Hasse diagram of minimal degenerations

EXAMPLES:

```
sage: from quivercombinatorics import *
sage: C = [[2, -1, 0], [-1, 2, -2], [0, -2, 2]]
sage: Q = quiver_from_cartan_matrix(C)
sage: Q.get_Hasse_diagram_method_1((0,0,0), (2,4,3))
```

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```
([[[(0, 0, 1), 1],
[(0, 1, 0), 2],
[(0, 1, 1), 1],
[(0, 1, 1), 1],
[(1, 0, 0), 2]],
[[(0, 0, 1), 1], [(0, 1, 0), 2], [(0, 1, 1), 2], [(1, 0, 0), 2]],
[[(0, 0, 1), 2], [(0, 1, 0), 3], [(0, 1, 1), 1], [(1, 0, 0), 2]],
[[(0, 0, 1), 3], [(0, 1, 0), 4], [(1, 0, 0), 2]],
[[(0, 1, 0), 1],
[(0, 1, 1), 1],
[(0, 1, 1), 1],
[(0, 1, 1), 1],
[(1, 0, 0), 2]],
[[(0, 1, 0), 1], [(0, 1, 1), 1], [(0, 1, 1), 2], [(1, 0, 0), 2]],
[[(0, 1, 0), 1], [(0, 1, 1), 3], [(1, 0, 0), 2]],
[[(2, 4, 3), 1]]],
[0, 1, 2, 3, 4, 5, 6, 7],
[(0, 4),
(1, 5),
(1, 0),
(2, 0),
(2, 5),
(3, 2),
(3, 1),
(3, 6),
(4, 7),
(5, 4),
(6, 5)],
[(0, 4, '$A_1(1)$'),
(1, 5, '$A_1(1)$'),
(1, 0, '$c_{1}(1)$'),
(2, 0, '$A_1(1)$'),
(2, 5, '$A_1(1)$'),
(3, 2, '$A_1(1)$'),
(3, 1, '$A_1(1)$'),
(3, 6, '$A_1(1)$'),
(4, 7, '$D_{4}(1)$'),
(5, 4, '$c_{1}(1)$'),
(6, 5, '$m_{1}(1)$')])
```

Quiver.**plot_Hasse_diagram_method_1**(*l*, *v*, *dimensions=False*)

Constructs the Hasse diagram using Method 1, using `get_Hasse_diagram_method_1`. The output can then be plotted by appending `.plot()` for example

INPUT:

- `l` – an element of $\mathbb{Z}Q_0$
- `v` – an element of $\mathbb{N}Q_0$
- `dimensions` – When set to `True`, the leaves will be labeled by (`symplectic_leaf_dimension`, `counter`), where `counter` enumerates symplectic leaves of equal dimensions

OUTPUT: The Hasse diagram as a poset

EXAMPLES:

```
sage: from quivercombinatorics import *
sage: C = [[2, -1, 0], [-1, 2, -2], [0, -2, 2]]
sage: Q = quiver_from_cartan_matrix(C)
sage: Q.plot_Hasse_diagram_method_1((0, 0, 0), (2, 4, 3))
Finite poset containing 8 elements

sage: Q = LoopQuiver(3)
sage: Q.plot_Hasse_diagram_method_1((0), (4), dimensions=True)
Finite poset containing 11 elements
```

To export this Hasse diagram as a TikZ diagram, we need to install `dot2tex` and `graphviz`.

```
sage --pip install dot2tex
sage --pip install graphviz
```

and then make sure you import the following

```
from sage.all import latex
```

`Quiver.`**`plot_Hasse_diagram_method_1_labels`**(*l*, *v*, *dimensions=False*, *latex=False*)

Constructs the Hasse diagram using Method 1, using `get_Hasse_diagram_method_1`, but also outputs the labels of the minimal degenerations. The TikZ code can be seen by applying `latex()` to the output when `latex` is set to `True`.

INPUT:

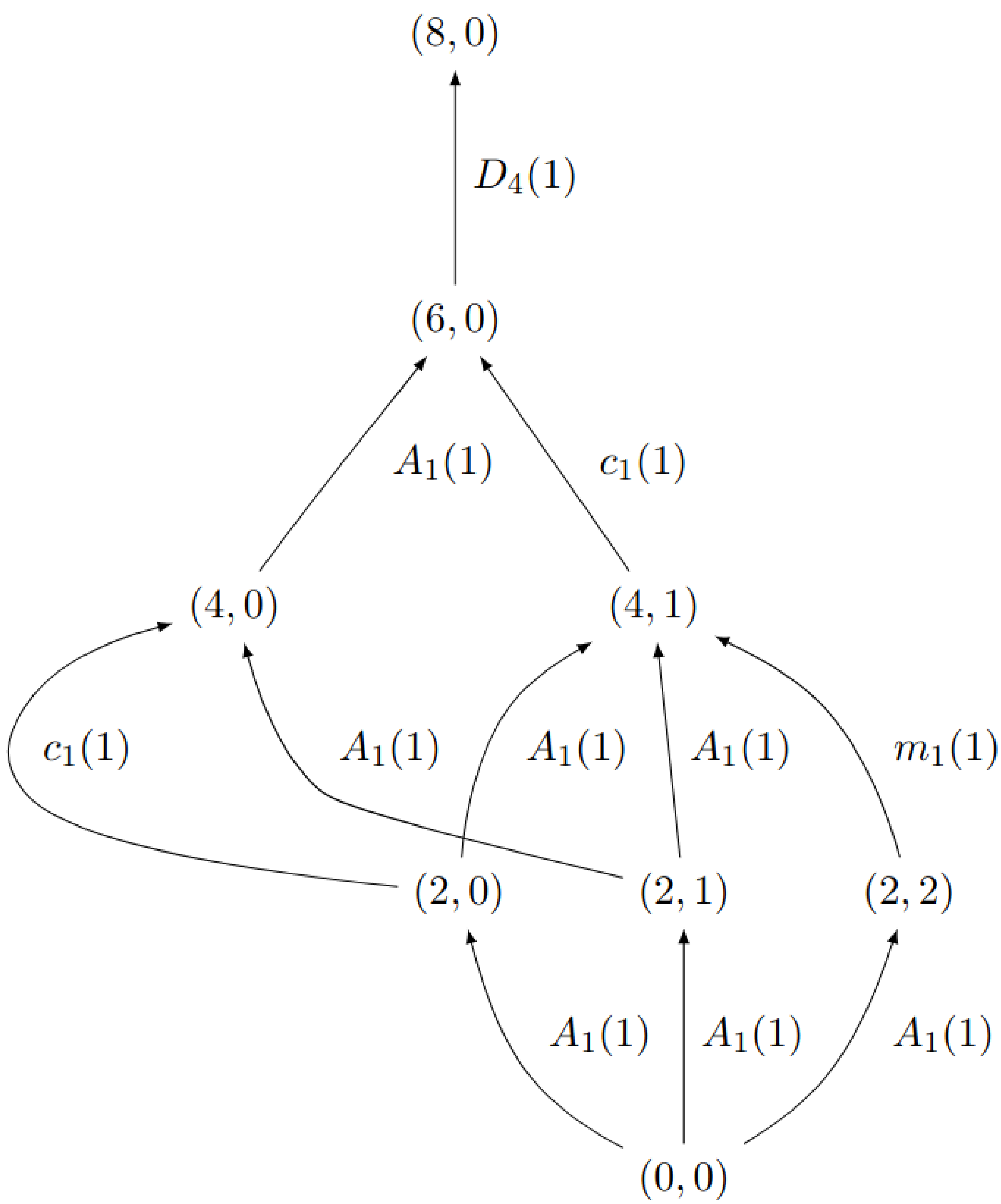
- `l` – an element of $\mathbb{Z}Q_0$
- `v` – an element of $\mathbb{N}Q_0$
- `dimensions` – When set to `True`, the leaves will be labeled by (`symplectic_leaf_dimension`, `counter`), where `counter` enumerates symplectic leaves of equal dimensions
- `latex` – When set to `True`, the labels will be formatted to be ready for a latex export

OUTPUT: The Hasse diagram as a poset, with labels

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: from sage.all import latex
sage: C = [[2, -1, 0], [-1, 2, -2], [0, -2, 2]]
sage: Q = quiver_from_cartan_matrix(C)
sage: P = Q.plot_Hasse_diagram_method_1_labels((0, 0, 0), (2, 4, 3), latex=True,
↪dimensions=True)
sage: P.set_latex_options(format='dot2tex', prog='dot', rankdir='up', edge_
↪labels=True, color_by_label=False)
sage: tex_code = latex(P)
sage: with open("poset_diagram.tex", "w") as f:
sage; f.write(tex_code)
```

The example above should export the following diagram:




The second method to compute this Hasse diagram is as follows.

**Method 2**

Consider the larger set $\mathcal{D}$ of all decompositions of a given dimension vector $v$. Here, a decomposition is simply a way of writing $v$ as a sum of smaller dimension vectors with multiplicities:

$$\left(n_1, \beta^{(1)}; \ldots; n_k, \beta^{(k)}\right),$$

where the $\beta^{(i)}$ are dimension vectors satisfying $\beta^{(i)} \leq v$. Following this paper, we say that:

$$\varepsilon = (p_1, \gamma^{(1)}; \dots; p_l, \gamma^{(l)}) > \rho = (m_1, \alpha^{(1)}; \dots; m_r, \alpha^{(r)})$$

is a *direct successor* ($\varepsilon$ is the successor of $\rho$) if either:

1. $l = r + 1$ and $\varepsilon$ can be ordered such that for all $1 \leq i \leq r - 1$ we have $(m_i, \alpha^{(i)}) = (p_i, \gamma^{(i)})$ and $m_r = p_r = p_{r+1}, \gamma^{(r)} + \gamma^{(r+1)} = \alpha^{(r)}$.
2. $l = r - 1$ and $\varepsilon$ can be ordered such that for all $1 \leq i \leq r - 2$ we have $(m_i, \alpha^{(i)}) = (p_i, \gamma^{(i)})$ and $\alpha^{(r)} = \alpha^{(r-1)} = \gamma^{(r-1)}, m_{r-1} + m_r = p_{r-1}$.

The symplectic leaves form a subset of this set of decompositions, and by restriction, they inherit a sub-poset structure. It is shown in (Bellamy-Schedler, 2026) that this is the partial ordering given by leaf closure.

`Quiver.`**`all_decompositions`**`(`*v*`)`

Constructs all decompositions of a given dimension vector $v$

INPUT:

- `v` – an element of $\mathbb{N}Q_0$

OUTPUT: A list of decompositions of *v*

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: A = [[0, 1, 0, 3], [1, 0, 0, 0], [0, 0, 2, 0], [0, 0, 0, 0]]
sage: Q = Quiver(A)
sage: Q.all_decompositions((1, 0, 2, 0))
[[[(0, 0, 1, 0), 1], [(0, 0, 1, 0), 1], [(1, 0, 0, 0), 1]],
[[(0, 0, 1, 0), 1], [(1, 0, 1, 0), 1]],
[[(0, 0, 1, 0), 2], [(1, 0, 0, 0), 1]],
[[(0, 0, 2, 0), 1], [(1, 0, 0, 0), 1]],
[[(1, 0, 2, 0), 1]]]
```

`quivercombinatorics.`**`is_direct_successor`**`(`*epsilon*, *rho*`)`

Verifies whether `epsilon` is the direct successor of `rho`

INPUT:

- `epsilon` – an element of $\mathbb{N}^n$
- `rho` – an element of $\mathbb{N}^n$

OUTPUT: `True` if `epsilon` is the direct successor of `rho`, and `False` otherwise

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: rho = [((1, 0), 2), ((1, 0), 3)]
sage: epsilon = [((1, 0), 5)]
sage: is_direct_successor(epsilon, rho)
True
```

`Quiver.`**`get_Hasse_diagram_method_2`**`(`*l*, *v*`)`

Applies Method 2 to obtain data for the Hasse diagram of minimal degenerations

INPUT:

- `l` – an element of $\mathbb{Z}Q_0$
- `v` – an element of $\mathbb{N}Q_0$
- `dimensions` – When set to `True`, the leaves will be labeled by (`symplectic_leaf_dimension`,``counter``), where `counter` enumerates symplectic leaves of equal dimensions

OUTPUT: `rep_types`, `dimensions`, `rep_types_poset`, the last output is the actual poset

EXAMPLES:

```
sage: from quivercombinatorics import *
sage: C = [[2, -1, 0], [-1, 2, -2], [0, -2, 2]]
sage: Q = quiver_from_cartan_matrix(C)
sage: Q.get_Hasse_diagram_method_2((0, 0, 0), (2, 4, 3))

([[[(0, 0, 1), 1],
[(0, 1, 0), 2],
[(0, 1, 1), 1],
[(0, 1, 1), 1],
[(1, 0, 0), 2]],
[[(0, 0, 1), 1], [(0, 1, 0), 2], [(0, 1, 1), 2], [(1, 0, 0), 2]],
[[(0, 0, 1), 2], [(0, 1, 0), 3], [(0, 1, 1), 1], [(1, 0, 0), 2]],
[[(0, 0, 1), 3], [(0, 1, 0), 4], [(1, 0, 0), 2]],
[[(0, 1, 0), 1],
[(0, 1, 1), 1],
[(0, 1, 1), 1],
[(0, 1, 1), 1],
[(1, 0, 0), 2]],
[[(0, 1, 0), 1], [(0, 1, 1), 1], [(0, 1, 1), 2], [(1, 0, 0), 2]],
[[(0, 1, 0), 1], [(0, 1, 1), 3], [(1, 0, 0), 2]],
[[(2, 4, 3), 1]]],
[4, 2, 2, 0, 6, 4, 2, 8],
Finite poset containing 8 elements)
```

`Quiver.`**`plot_Hasse_diagram_method_2`**($l$, $v$)

Constructs the Hasse diagram using Method 2, using `get_Hasse_diagram_method_2`. The output can then be plotted by appending `.plot()` for example

INPUT:

- `l` – an element of $\mathbb{Z}Q_0$
- `v` – an element of $\mathbb{N}Q_0$
- `dimensions` – When set to `True`, the leaves will be labeled by (`symplectic_leaf_dimension`, `counter`), where `counter` enumerates symplectic leaves of equal dimensions

OUTPUT: The Hasse diagram as a poset

EXAMPLES:

```
sage: from quivercombinatorics import *
sage: C = [[2, -1, 0], [-1, 2, -2], [0, -2, 2]]
sage: Q = quiver_from_cartan_matrix(C)
sage: Q.plot_Hasse_diagram_method_2((0, 0, 0), (2, 4, 3))
Finite poset containing 8 elements
```

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```
sage: Q = LoopQuiver(3)
sage: Q.plot_Hasse_diagram_method_2((0), (4), dimensions=True)
Finite poset containing 11 elements
```

It seems that Method 2 is slower than Method 1, so we only implemented a TikZ exporter for Method 1.